MATH329 · OPERATIONS RESEARCH

Midterm 472 Solutions

13/04/2026 · 60 minutes · 30 marks

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01 Solution Q1: True / False

#	Answer	Reason
1	**F**	The graphical method works for two variables only, not for more than three.
2	**F**	It started during **World War II**, not the first war.
3	**F**	The objective and all the constraints must be linear, not one or the other.
4	**T**	The tableau organizes the objective and the constraint equations.
5	**F**	The solution must satisfy the constraints, not "may".
6	**T**	This is one of the two requirements to start simplex.
7	**T** *	This is what the answer key shows. See the note below.

A note on item 7

The key shows T. By the lecture, when no variable can enter, the current solution is optimal (**the last solution is optimal**); it does not mean the problem has no feasible solution. So the logical answer is F. Ask the instructor before the midterm, and do not rely on my answer or on the key.

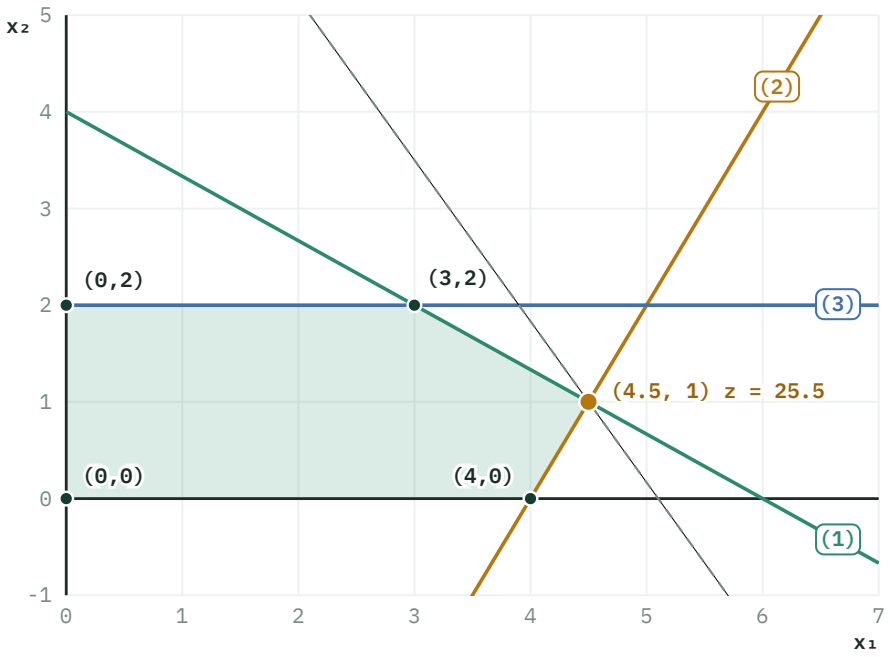
The answer to item 2 is not visible in the photo of the key; F is taken from the lecture.

02 Solution Q2: graphical

1 Prepare the constraints

(1)	$2x_1 + 3x_2 = 12$	points (0, 4) and (6, 0)
(2)	$2x_1 - x_2 = 8$	points (4, 0) and (0, −8)
(3)	$-2x_2 \geq -4$ × (−1) →	$x_2 \leq 2$ horizontal line at $x_2 = 2$

Testing `(0, 0)` satisfies all three, so the region is on the origin's side.



2 The corner points

Corner	Where it comes from	$Z = 5x_1 + 3x_2$
(0, 0)	origin	0
(4, 0)	(2) with the x_1 axis	20
(4.5, 1)	(1) with (2)	**25.5**
(3, 2)	(1) with (3)	21
(0, 2)	(3) with the x_2 axis	6

(1) − (2): $4x_2 = 4 \rightarrow x_2 = 1$ $\quad$ $2x_1 - 1 = 8 \rightarrow x_1 = 4.5$

Objective Value = 25.5 at $x_1 = 4.5$ and $x_2 = 1$

Marks in the key: one for each of the three lines, one for the objective line, two for the solution space, one for the optimum point, and two for the answer.

03 Solution Q3: standard form and simplex

a Standard form 5 M

The second constraint has a negative right-hand side, so multiply it by -1 first:

$$x_1 - x_2 \geq -1 \quad \times (-1) \quad \rightarrow \quad -x_1 + x_2 \leq 1$$

$$\begin{aligned}
&\text{Maximize } Z = 16x_1 + 15x_2 + 0s_1 + 0s_2 + 0s_3 \\
&\text{Subject to} \quad 40x_1 + 31x_2 + s_1 = 124 \\
&\qquad\qquad -x_1 + x_2 + s_2 = 1 \\
&\qquad\qquad x_1 + s_3 = 3 \\
&\qquad\qquad x_1, x_2, s_1, s_2, s_3 \geq 0
\end{aligned}$$

b **Simplex** 10 M · 2 TABLES ONLY

Basic	x_1	x_2	s_1	s_2	s_3	Sol.	Ratio
Z	−16	−15	0	0	0	0	
s_1	40	31	1	0	0	124	$\frac{31}{10}$
s_2	−1	1	0	1	0	1	–
s_3	**1**	0	0	0	1	3	3
Z	0	−15	0	0	16	48	
s_1	0	31	1	0	−40	4	
s_2	0	1	0	1	1	4	
x_1	1	0	0	0	1	3	

The most negative entry in the `Z` row is `−16`, so x_1 enters. The ratios are $\frac{124}{40} = 3.1$ for the first row, none for the second row (negative coefficient), and $\frac{3}{1} = 3$ for the third. The smallest is `3`, so s_3 leaves and the pivot is `1`.

New x_1 row = s_3 row ÷ 1 = (1, 0, 0, 0, 1 | 3)

New Z row = Z row − (−16) × new x_1 row = (0, −15, 0, 0, 16 | 48)

New s_1 row = s_1 row − 40 × new x_1 row = (0, 31, 1, 0, −40 | 4)

New s_2 row = s_2 row − (−1) × new x_1 row = (0, 1, 0, 1, 1 | 4)

We stop here because the question asks for two tables only. The solution is not optimal yet, because `−15` remains in the `Z` row.